DO NOW — RECALL

OOP: testing and debugging

Name: _____ Class: _____ Date: _____

1. Insert 12 into a BST node 10 with empty right child. Where does it go?

2. What is a reference?

3. What is a local variable?

4. If an object's Balance=20 and Deposit(5) correctly adds five, what is Balance?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

As the right child of 10.

2. Model answer

A value that identifies an object rather than copying the object itself.

3. Model answer

A variable declared inside a method or block with limited scope.

4. Model answer

25.

Vocabulary — know these terms

encapsulation — Hide implementation details and control state through methods/properties.

constructor — Initialises a new object.

field / property — Stores or exposes object state.

breakpoint — Pauses execution at a selected code line.

boundary test — Checks values at or just beyond a permitted limit.

NEW LEARNING

Read, annotate and explain

Inspect class state

The object has a Balance property initially set by its constructor.

Trace method inputs

Deposit(5) passes amount=5. Since amount>0, Balance+=5 executes.

Debug the outcome

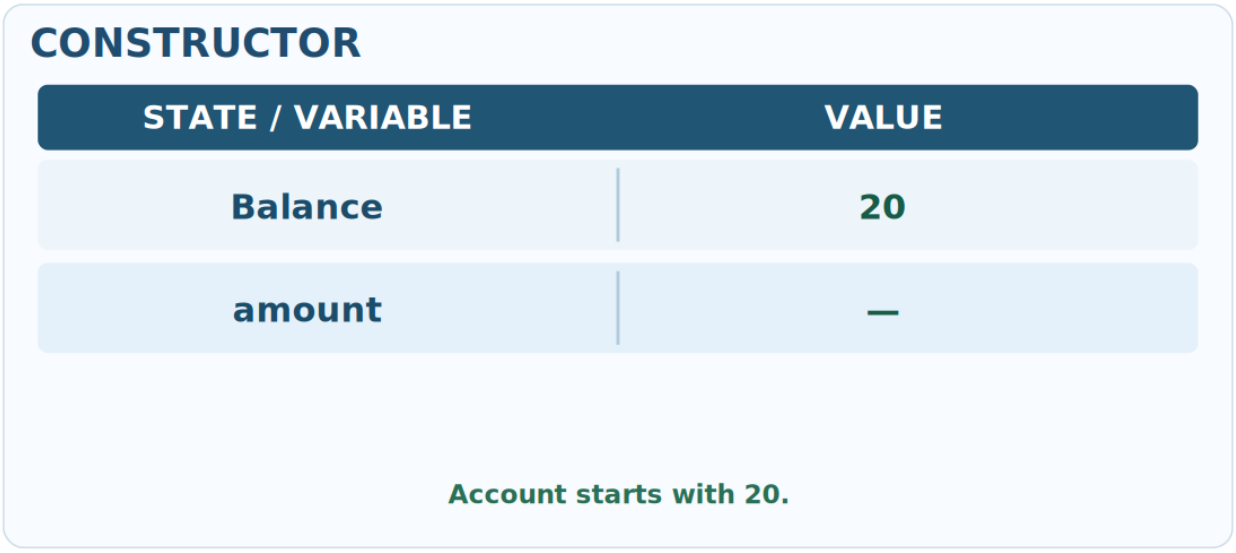
Expected Balance=25. If unchanged, check whether the code assigns to Balance or merely calculates a temporary value.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

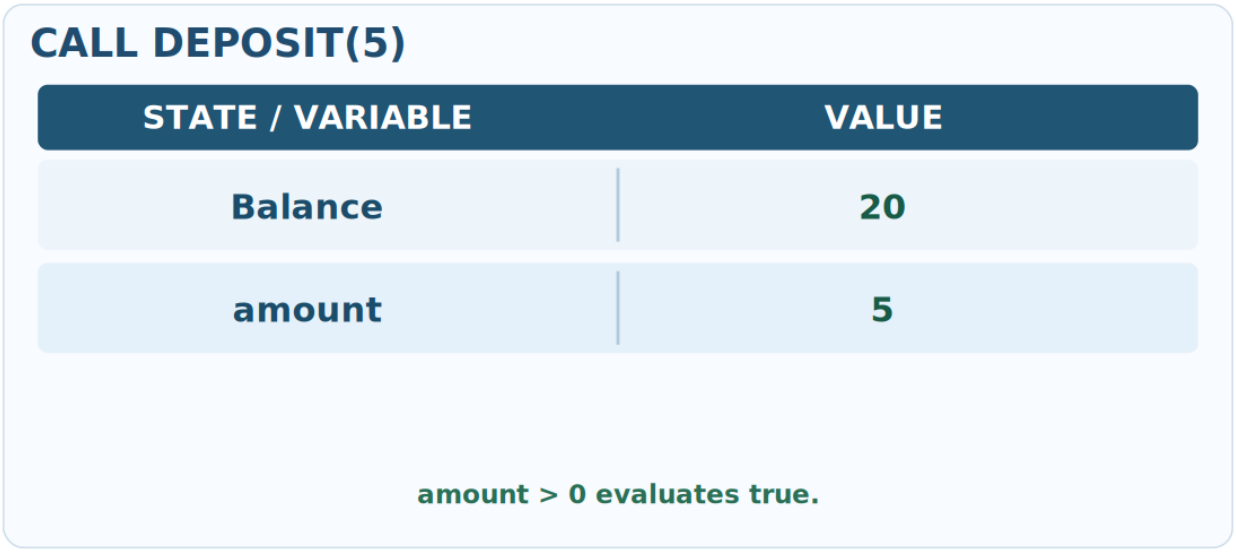
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CONSTRUCTOR Balance equals 20; amount equals —. Account starts with 20.

New Learning

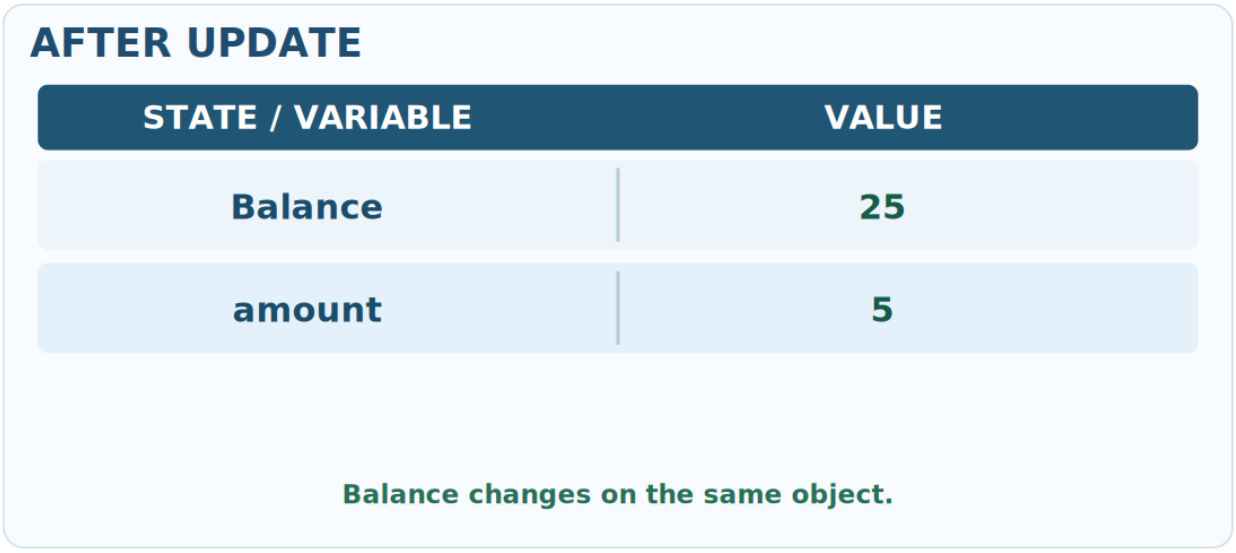


CALL DEPOSIT(5) Balance equals 20; amount equals 5. amount > 0 evaluates true.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



AFTER UPDATE Balance equals 25; amount equals 5. Balance changes on the same object.

I DO — WORKED EXAMPLE

Follow the worked method

Find and correct the fault when AddPoints(3) fails to update Score=7.

Predict first

The local newScore becomes 10, but the field Score remains 7.

Model the reasoning

Replace the local-only calculation with Score += points;.

Model the reasoning

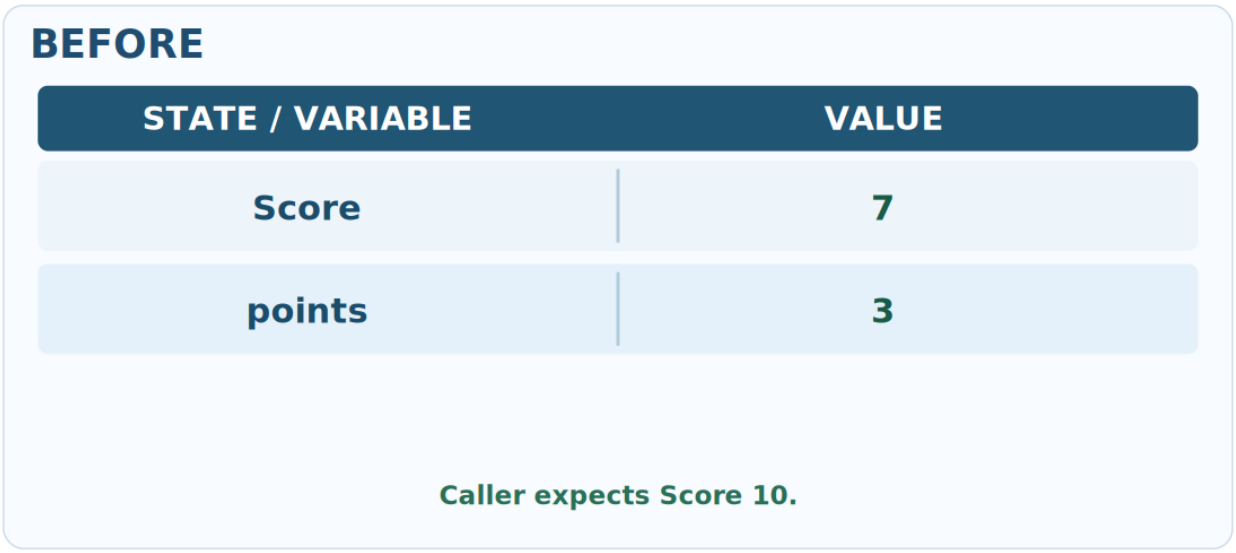
Retest: after AddPoints(3), Score should equal 10.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find and correct the fault when AddPoints(3) fails to update Score=7.

The local newScore becomes 10, but the field Score remains 7.



BEFORE Score equals 7; points equals 3. Caller expects Score 10.

Find and correct the fault when AddPoints(3) fails to update Score=7.

The local newScore becomes 10, but the field Score remains 7.



FAULT newScore equals 10; Score equals 7. Calculated but never stored.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find and correct the fault when AddPoints(3) fails to update Score=7.

The local newScore becomes 10, but the field Score remains 7.



FIX Score equals 10; instruction equals Score += points. Now object state is updated.

WE DO — GUIDED PRACTICE

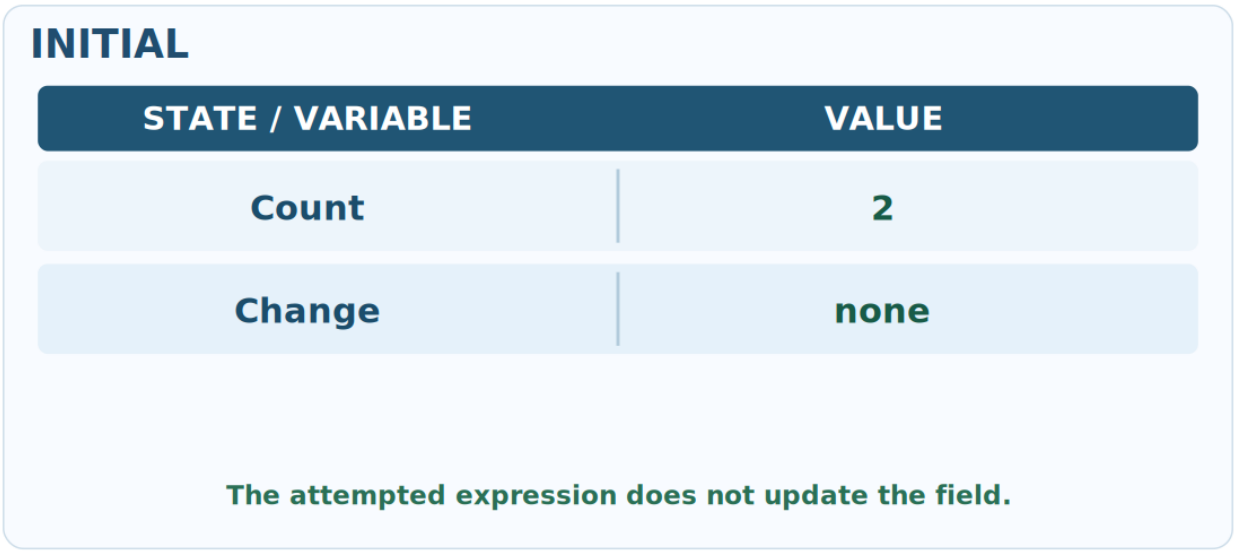
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Counter starts at 2. Increment() contains `Count + 1;`. Explain the problem and fix it.

Count + 1 does not assign a new value and is not a valid standalone statement in C#.



INITIAL Count equals 2; Change equals none. The attempted expression does not update the field.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Count + 1 does not assign a new value and is not a valid standalone statement in C#.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

An object starts at Count=4 and runs Increment() which performs Count += 1. What is Count after three calls?

Question 2

Correct `public void SetAge(int age) { int Age = age; }` where Age is an existing property. Explain why it fails.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A BankAccount.Withdraw(amount) rejects only amounts <=0, but permits overdrafts. Add validation and give boundary tests for Balance=50.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A method may calculate a correct new value without actually updating the object's property.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

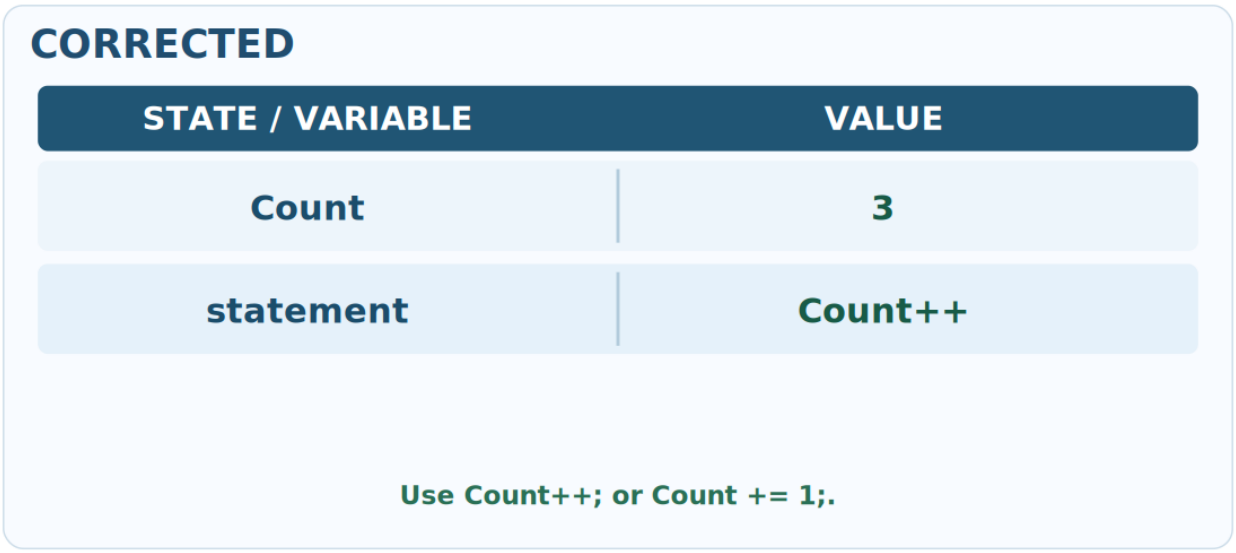
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Counter starts at 2. Increment() contains `Count + 1;`. Explain the problem and fix it.

Count + 1 does not assign a new value and is not a valid standalone statement in C#.



CORRECTED Count equals 3; statement equals Count++. Use Count++; or Count += 1;.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model the reasoning

Correct it to Count++; or Count += 1;. With Count=2, the resulting Count is 3.

IMPROVE: Explain one correction to your own reasoning.